1968 Penrose argues that it is impossible to travel from the interior of a black hole in our Universe through hyperspace and into some other universe; others, in the 1970s, will confirm that his argument is correct. [Ch. 13]

Carter discovers the nature of the swirl of space around a spinning black hole and its influence on infalling particles. [Ch. 7]

Misner and independently Belinsky, Khalatnikov, and Lifshitz discover the oscillatory "mixmaster" singularity as a solution of Einstein's equation. [Ch. 13]

1969 Hawking and Penrose prove that our Universe must have had a singularity at the beginning of its big bang expansion. [Ch. 13]

Belinsky, Khalatnikov, and Lifshitz discover the oscillatory BKL singularity as a solution of Einstein's equation; they show that it has random deformations of its spacetime curvature and argue that therefore it is the type of singularity that forms inside black holes and in the big crunch. [Ch. 13]

Penrose discovers that a spinning black hole stores enormous energy in the swirling motion of space around it, and that this rotational energy can be extracted. [Ch. 7]

Penrose proposes his cosmic censorship conjecture, that the laws of physics prevent naked singularities from forming. [Ch. 13]

Lynden-Bell proposes that gigantic black holes reside in the nuclei of galaxies and are surrounded by accretion disks. [Ch. 9]

Christodoulou notices a similarity between the evolution of a black hole when it slowly accretes matter and the laws of thermodynamics. [Ch. 12]

Weber announces tentative observational evidence for the existence of gravitational waves, triggering many other